

When Difficulty Signals Are Unreliable: Auditing and Gating Sample-Count Adaptive Test-Time Compute

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Abstract. Adaptive test-time compute (TTC) can reduce sample cost by allocating extra decodes to inputs predicted to benefit from them, but this requires a signal that ranks marginal compute utility reliably after deployment. We audit a diagnostic binary allocator whose ANLL threshold is frozen on a source model and reused without target labels. Across Qwen3 open-weight models from 1.7B to 30B on MATH-500, ANLL allocation falls below uniform in 13 of 20 cross-model transfers. Off-diagonal means vary more systematically by deployment target than by calibration source, so model transfer exposes the instability but is not its sole cause. We evaluate METAGATE, a label-free fallback that trusts the allocator when ANLL and answer consistency agree and otherwise uses uniform allocation. With the predeclared setting $k = 4, \alpha = 2.0, \tau = 0.5$, METAGATE uses 6.99 decoded samples per query and averages +0.28 pp over uniform (query-clustered 95% CI [+0.05, +0.52]), with one observed below-uniform transfer. It also beats a sample-budget-matched random fallback by +0.79 pp. Consistency is a strong standalone allocator (+0.66 pp at 8.33 samples); applying the unchanged gate retains +0.31 pp at 7.20 samples with the same downside count. These results establish two useful operating points: higher accuracy from consistency and lower decoded-sample cost from gated consistency. An additional Phi-4-mini-instruct run provides cross-family corroboration of the gate's selectivity.

Keywords: Large language models · Test-time compute · Adaptive inference · Difficulty estimation · Model transfer · Selective prediction.

1 Introduction

For reasoning tasks, repeated decoding often improves majority-vote accuracy. Sampling many chains and aggregating by majority vote [30,4], generating longer chains of thought [31], and training models to deliberate [11] all trade test-time compute (TTC) for accuracy, often more favorably than scaling the model itself [26,32]. The cost, however, is asymmetric: generating N trajectories multiplies inference cost N -fold, yet many queries are already correct with one. This

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has motivated *adaptive* TTC, which spends more compute on hard inputs and less on easy ones.

Adaptive TTC methods change either the amount of decoding or the model/strategy used per query. They typically rely on an estimated *difficulty signal* $s(x)$ that ranks inputs by how much compute they warrant. The allocator is only as good as this signal: when s separates easy from hard, reallocation helps; when it does not, compute is misdirected.

Deployment without recalibration. The adaptive-TTC literature typically estimates and consumes a difficulty signal under the same model and distribution. Deployment can instead reuse a frozen budget rule after the backbone changes, when target labels are unavailable. This setting raises two distinct questions: whether the source calibration transfers, and whether the signal ranks marginal compute utility well on the deployment model at all. Existing risk-control, routing, and cascade methods usually assume some calibration condition. Here the label-free decision is therefore whether a deployed allocator should be trusted or replaced by a uniform fallback.

Our study. We study allocator reliability under cross-model deployment without target-label recalibration. First, we use ANLL in a diagnostic binary source-quantile allocator and audit the full source-to-target matrix across five Qwen3 models. The binary allocator is our controlled test mechanism, distinct from the graded allocation rule in the ANLL work we cite. Second, we evaluate META-GATE, which uses agreement between ANLL and answer consistency to select the frozen allocator or a uniform fallback. Third, we compare the unchanged gate with stronger consistency and entropy base allocators. The results separate three effects that the deployment setting brings together: deployment-target dependence, base-signal quality, and the gate’s cost–downside tradeoff. We also include a Phi-4-mini-instruct check for cross-family corroboration.

The resulting practical question is not simply whether model transfer is harmful. It is when a frozen allocator remains useful on the deployed model, and when a label-free policy should return to a uniform budget.

2 Related Work

2.1 Adaptive Test-Time Compute

Repeated sampling and inference scaling show why per-query compute allocation matters [4,32,26]. One line of adaptive TTC varies the decoding budget through repeated sampling or early stopping [29,2,16,19]. Another uses budget-conditioned reasoning length or difficulty-aware budgeting [21,35,13,18,15,38]. In particular, the ANLL method [21] uses a global temperature and a graded marginal-greedy allocation rule, whereas our controlled audit uses a per-source quantile threshold. We adopt its low-cost ANLL signal and introduce the binary quantile allocator as a controlled audit mechanism. Related routing work chooses

among models, strategies, or learned allocation rules [23,14,6,36,24]. Most evaluations estimate and consume the difficulty signal under the same model and data setting; we study the no-recalibration case where the backbone changes and target labels are unavailable.

2.2 Risk Control and Robustness

Conformal risk control chooses thresholds to control expected risk or error against a user tolerance [28,3], with additional assumptions under covariate shift [27]. Distributionally robust routing hedges against an assumed shift set [37]. Our setting differs: after a backbone change, we test whether the transferred signal remains reliable without target labels. We focus on a deployable check for whether to use the allocator, with an exact selective-risk characterization of the gated policy.

2.3 Brittleness of Difficulty Signals

Probes trained on pre-generation activations lose discriminative power as reasoning budgets grow [20]. Solvability-aware work shows nominal difficulty and actual ability to solve diverge [38], and reasoning accuracy itself can collapse non-monotonically with problem complexity [25]; neural-network confidence is often miscalibrated under shift [10,22]. These failures motivate checking the allocator signal itself, separately from final-answer confidence. METAGATE is related to selective prediction [8], but the gate still returns an answer; it only replaces adaptive allocation with uniform allocation. It also uses uncertainty and disagreement diagnostics [9,33,7], adapted here to allocator-input reliability.

3 Problem Formulation

3.1 Adaptive Allocation and the Difficulty Signal

Let f be a language model and x an input query. An adaptive TTC method chooses a per-query budget $b \in \mathcal{B} = \{b_1 < \dots < b_K\}$ and runs f at that budget. Let $\text{Acc}(x, b) \in [0, 1]$ denote expected correctness under budget b over sampling randomness and answer aggregation, and $C(b)$ its compute cost. Given a target average budget B , the idealized allocation is

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}} [\text{Acc}(x, \pi(x))] \quad \text{s.t.} \quad \mathbb{E}_{x \sim \mathcal{D}} [C(\pi(x))] \leq B, \quad (1)$$

where $\pi : x \mapsto b$. Solving (1) exactly requires the per-query accuracy profile, which is unknown. Many practical methods therefore use a low-cost *difficulty signal* $s(x) \in \mathbb{R}$ as a proxy: low s marks easy inputs, high s marks hard ones. The policy factorizes as $\pi(x) = g(s(x))$ for some monotone map g tuned on a source model f_{src} and source distribution $\mathcal{D}_{\text{src}}$.

3.2 Cross-Model Deployment Without Recalibration

Many adaptive-TTC evaluations fit g under the same model-distribution pair used for evaluation; we consider deployment on $(f_{\text{tgt}}, \mathcal{D}_{\text{tgt}})$ with no labels to re-fit g . The signal is useful only if it preserves the ordering induced by marginal compute utility:

$$\rho_{\text{tgt}} = \text{Spearman}_{x \sim \mathcal{D}_{\text{tgt}}}(s(x), \Delta\text{Acc}(x)), \quad \Delta\text{Acc}(x) = \text{Acc}(x, b_K) - \text{Acc}(x, b_1). \quad (2)$$

Here ΔAcc is an oracle measure of marginal utility from extra test-time samples, not an intrinsic semantic difficulty score. Our implementation uses the binary adaptive case with $b_1 = k$ and $b_K = N$. In deployment, the budget rule is calibrated on the source distribution; on the target distribution we monitor and report realized budget. When ρ_{tgt} is low, the allocator can be worse than uniform because it withholds compute from inputs that needed it, while mis-ranked easy inputs mainly create opportunity cost.

3.3 Meta-Confidence for Signal Reliability

We introduce the *meta-confidence* $m(x) \in [0, 1]$, a label-free proxy for the local reliability of $s(x)$. The deployment policy selects between adaptive allocation and a uniform-budget fallback π_0 . For sample-count budgets, we use the same integer B as the uniform per-query fallback:

$$\pi_{\text{gate}}(x) = \begin{cases} g(s(x)), & m(x) \geq \tau, \\ \pi_0(x), & m(x) < \tau, \end{cases} \quad (3)$$

where τ controls downside risk. A useful meta-confidence should require no target labels or target-side fitting. It should also track allocation errors after a model change.

4 METAGATE: Meta-Confidence Gating for Adaptive TTC

At deployment, METAGATE computes two signals from the probing decodes: log-probability and answer consistency. If the two signals agree, it uses the source-calibrated adaptive allocator. If they disagree, it uses uniform allocation. Figure 1 shows how the frozen source statistics and threshold enter the target-side decision.

4.1 Observable Signals at Deployment

The base allocator consumes a difficulty signal $s(x)$ calibrated on $(f_{\text{src}}, \mathcal{D}_{\text{src}})$; unless stated otherwise, this is ANLL s_{lp} . For a first probing completion $y^{(1)}$ from the target model,

$$s_{\text{lp}}(x) = -|y^{(1)}|^{-1} \sum_t \log p(y_t^{(1)} \mid y_{<t}^{(1)}, x),$$

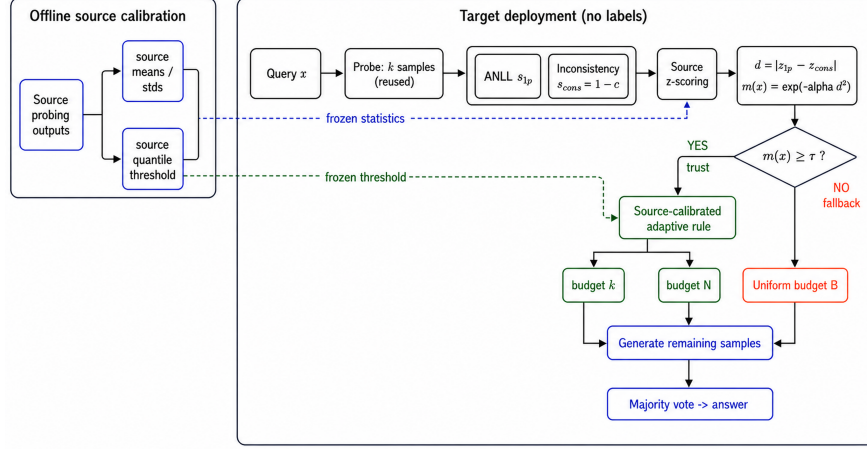


Fig. 1. METAGATE pipeline. Source statistics and the allocator threshold are frozen after offline calibration. At deployment, the same k probing samples supply both signals and are reused in the selected k , B , or N -sample vote. Dashed arrows carry frozen parameters.

so larger values mean lower model confidence. We also generate k probing samples (default $k = 4$) and compute answer consistency: if a^* is the majority extracted answer, then $c(x) = k^{-1} \sum_i \mathbf{1}[a_i = a^*]$ and $s_{\text{cons}}(x) = 1 - c(x)$. The same deterministic extraction and tie-breaking rules are used for both the signal and final evaluation.

ANLL depends on the target model’s probability scale; consistency depends on the sampled answer distribution. Their disagreement is useful because common failures leave different traces. A model can assign low token confidence to otherwise stable answers, or it can be confident while the sampled answers split across alternatives. We use agreement between the two signals as a deterministic meta-confidence statistic for allocator selection. This is related to self-consistency and convergence methods [30,16,19] and uncertainty diagnostics [9,33], but the object being judged is allocator reliability rather than final-answer correctness.

4.2 Meta-Confidence and Gate

Let $\hat{s}_{\text{IP}}$ and $\hat{s}_{\text{cons}}$ be z-scored with source-calibration statistics $(\mu_{\text{IP}}, \sigma_{\text{IP}})$ and $(\mu_{\text{cons}}, \sigma_{\text{cons}})$, estimated once on $\mathcal{D}_{\text{src}}$ and fixed at deployment. This avoids target-side statistics. With an implementation floor on source standard deviations, define

$$d(x) = |\hat{s}_{\text{IP}}(x) - \hat{s}_{\text{cons}}(x)|, \quad m(x) = \exp[-\alpha d(x)^2]. \quad (4)$$

Small discrepancy gives high meta-confidence; large disagreement drives m toward zero. Before the transfer evaluation, we fixed $k = 4$, $\alpha = 2.0$, and $\tau = 0.5$

without target-label tuning; nearby values are swept only as sensitivity analyses. This threshold accepts discrepancies up to $\sqrt{\log 2/\alpha} \approx 0.59$ in source-z-score units. Since $m \in (0, 1]$, $\tau \leq 0$ recovers the base adaptive allocator and $\tau > 1$ recovers uniform allocation.

Budget accounting. The first k probing samples are counted in the decoded-sample budget and reused in final majority vote, so the adaptive minimum budget is k rather than 1. In our sample-budget setting $C(b) = b$, with integer $k \leq B \leq N$; the fallback branch uses uniform budget B . The source allocator sends fraction $f = (B - k)/(N - k)$ to N and the rest to k , matching budget B only on the source calibration distribution. Under transfer, the trusted subset can have a different high-budget fraction, so we report realized budgets and compare against sample-budget-matched random fallback.

4.3 Selective Risk Accounting

Let $A_\tau(x) = \mathbf{1}[m(x) \geq \tau]$, $\pi_{\text{adp}}(x) = g(s(x))$, and let π_τ denote the gated policy in Eq. (3). Define $L(x, \pi) = 1 - \text{Acc}(x, \pi(x))$, $R(\pi) = \mathbb{E}[L(x, \pi)]$, and $q_\tau = \Pr(A_\tau = 1)$. Direct expansion of the two gate branches gives

$$R(\pi_0) - R(\pi_\tau) = q_\tau \mathbb{E}[L(x, \pi_0) - L(x, \pi_{\text{adp}}) \mid A_\tau = 1], \quad (5)$$

$$R(\pi_{\text{adp}}) - R(\pi_\tau) = (1 - q_\tau) \mathbb{E}[L(x, \pi_{\text{adp}}) - L(x, \pi_0) \mid A_\tau = 0]. \quad (6)$$

Selective improvement criterion. Equations (5)–(6) show that METAGATE improves on uniform whenever adaptive allocation has nonnegative conditional advantage on the trusted subset, and improves on the ungated allocator whenever uniform has nonnegative conditional advantage on the fallback subset. If both conditions hold, the gated policy weakly dominates both references in expected risk. The identities follow by substituting $L(x, \pi_\tau) = A_\tau L(x, \pi_{\text{adp}}) + (1 - A_\tau)L(x, \pi_0)$ and taking expectations. Our harmful-decision ranking, risk-coverage curves, and matched-budget controls in Sec. 5.4 measure this selectivity directly.

4.4 Implementation Summary

For each query, METAGATE generates k probing samples, computes ANLL, consistency, and m , then either applies the source-calibrated adaptive threshold or adds $B - k$ samples for uniform fallback. If the trusted adaptive branch chooses high budget, it adds $N - k$ samples; if it chooses low budget, it stops after probing. Every reported answer is a majority vote over the realized sample count. For alternative base signals, we keep the meta-confidence and gate fixed and replace only the adaptive threshold in the trusted branch by the chosen source-z-scored signal and source threshold. This isolates whether gating complements a stronger allocator without introducing another tuned gate.

5 Experiments

The experiments focus on the deployment question: when a source-calibrated allocator is moved to another model, does it beat a uniform sample budget, and can a label-free gate reduce the cases where it does not?

5.1 Experimental Setup

Models. We use Qwen3-1.7B, 4B, 8B, 14B, and 30B-A3B [34] in non-thinking mode with temperature 0.7 and top- $p=0.95$; 30B-A3B adds MoE architectural diversity. We also run Phi-4-mini-instruct [1] on MATH-500, creating non-Qwen source-target transfers with a distinct generation profile.

Benchmarks. We evaluate GSM8K [5] (1,319 grade-school math problems) and MATH-500 [17,12] (500 competition-math problems). GSM8K is near ceiling for these models, so MATH-500 is the main transfer-audit setting.

Signals and allocator. The main gate uses ANLL [21] and probing consistency $(1 - c(x))$ [30]. The adaptive allocator sends the top source-quantile fraction $f = (B - k)/(N - k)$ to $N = 16$ samples and the rest to $k = 4$, giving average budget $B=8$ only on the calibration distribution. This binary quantile rule is our diagnostic allocator; the cited ANLL method instead uses a global temperature and a graded marginal-greedy rule. Probing samples are reused in final majority vote, all reported budgets include them, and accuracies are recomputed deterministically from the stored decodes.

Cross-model transfer protocol. Source quantile thresholds and z-scoring statistics are computed once on the source model and frozen. Targets use the same signal, source z-scoring, and source threshold, with no target labels or target-side fits; labels enter only for reporting and statistical tests. Budgets are measured as decoded sample counts. Unless stated otherwise, source statistics use unlabeled source outputs on the same benchmark query set; Sec. 5.5 checks held-out source splits. Self-baseline means source=target.

Baselines and references. **Uniform** uses $B = 8$; **Adaptive** is the transferred ANLL threshold allocator; **Random-fallback** matches either fallback fraction or realized sample budget; and **Acc@16** is an always-max reference. MetaGate fixes $k = 4, \alpha = 2.0, \tau = 0.5$, chosen before the transfer evaluation, unless stated otherwise.

Reproducibility. We use vLLM with 16 decodes/query and a 1024-token cap, a chat prompt requesting a final `\boxed{}` answer, deterministic last-boxed extraction, and a symbolic/numeric verifier applied uniformly to all policies, budgets, and models. Majority-vote ties follow deterministic stored-answer order; MATH-500 uses all 500 items with no per-item tuning.

5.2 Experiment 1: Transferability Audit

ANLL has modest alignment with marginal compute utility in-distribution: its Spearman correlation with ΔAcc is between $\rho \approx 0.12$ and 0.14 (bootstrap 95% CIs, e.g. 8B: $[0.04, 0.21]$; 14B: $[0.06, 0.21]$; 30B: $[0.03, 0.20]$). Previously reported ANLL–correctness correlations measure a different target: allocation requires ranking marginal gain from additional samples. Figure 2 reports the full 5×5 matrices of policy–uniform accuracy on MATH-500. ANLL comes from ordinary decoding, making this deployment audit directly relevant to adaptive TTC. Section 5.5 compares stronger base signals under the same protocol.

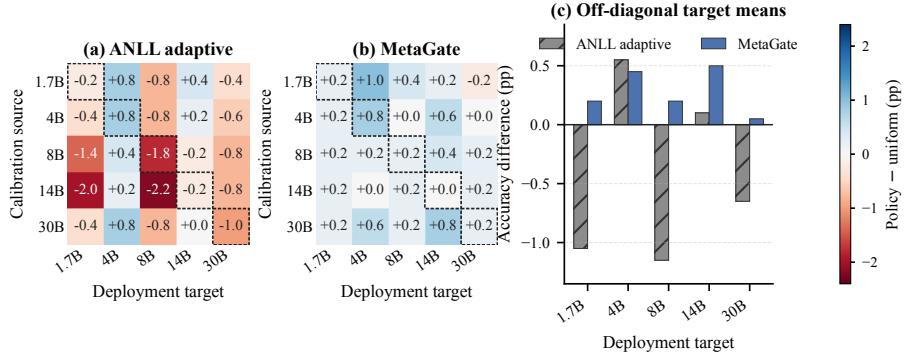


Fig. 2. MATH-500 source→target audit at $B=8$ and $k=4$. (a,b) Accuracy difference from uniform; dashed cells are self-baselines. (c) Off-diagonal means grouped by deployment target. Numeric labels and hatching preserve the comparisons in grayscale.

Across the 20 off-diagonal transfer cells, 13 are negative. Positive differences reach $+0.8$ pp, while the largest loss is -2.2 pp ($14B \rightarrow 8B$). Only $14B \rightarrow 8B$ reaches raw significance (paired McNemar $p=0.013$), and no cell survives Holm correction over 20 pairs. Transferred ANLL therefore does not provide a stable advantage over uniform. The realized adaptive budget under source-quantile transfer averages 8.03 samples (range 5.94 to 10.29), separating allocation quality from realized-budget variation. The similar diagonal and off-diagonal means (-0.48 and -0.44 pp) further localize the main effect to the deployment target.

Source versus deployment-target effects. Reading the matrix by rows, off-diagonal source means range from -1.20 to $+0.00$ pp. Reading by columns, deployment-target means range from -1.15 pp (8B) to $+0.55$ pp (4B): every source is positive on 4B, whereas every source is negative on 1.7B, 8B, and 30B. Thus transfer is not the active ingredient by itself; ANLL allocation helps or hurts primarily according to the deployed model, with the calibration source a secondary term. This target-dependent behavior motivates a label-free fallback path at deployment.

5.3 Experiment 2: METAGATE Evaluation

Figure 2b evaluates the fixed gate on the same source→target matrix. Unlike the ANLL allocator, METAGATE has only one observed below-uniform cell in the full matrix, and that cell is -0.2 pp. Across the 20 off-diagonal transfers, it averages $+0.28$ pp over uniform (query-clustered bootstrap 95% CI $[+0.05, +0.52]$), trusts 49.5% of queries, and uses 6.99 decoded samples per query (full-matrix range 6.58 to 7.57). On the 13 transfers where ANLL adaptive allocation is below uniform, METAGATE averages $+0.17$ pp over uniform at $\bar{b} = 6.98$ samples. These numbers quantify an empirical reduction in downside. Across the matrix, the gate largely recovers the uniform reference when the transferred signal is unreliable and retains some adaptive upside when the trusted subset is useful. Because one MATH-500 item is 0.2 pp, the query-clustered matrix summary is the primary measure of downside control and sample-budget reduction. The selective criterion in Eqs. (5)–(6) also yields a direct interpretation. With trust coverage $q_\tau = 0.495$, the $+0.28$ pp gain over uniform implies a $+0.28/0.495 = +0.57$ pp adaptive advantage on the trusted subset. Relative to ungated ANLL (-0.44 pp), the gate gains $+0.72$ pp overall, corresponding to a $+0.72/0.505 = +1.43$ pp uniform advantage on the fallback subset. Thus both selective-improvement conditions hold on the aggregate transfer distribution.

Table 1 places this operating point on a decoded-sample cost versus accuracy frontier. Under a query-clustered bootstrap that preserves each query’s repeated use across the 20 transfers, METAGATE($\tau=0.5$) exceeds uniform@7 by $+0.68$ pp with 95% CI $[+0.19, +1.22]$; its difference from uniform@10 is -0.08 pp with CI $[-0.53, +0.36]$. Nearby thresholds behave similarly: $\tau=0.3$ to 0.7 gives 0 or 1 degrading transfers and $+0.23$ to $+0.33$ pp over uniform@8; $\tau > 1$ recovers uniform.

Table 1. Cost versus accuracy frontier on the 20 off-diagonal MATH-500 transfers. Accuracy is macro-averaged over source→target pairs. Δ is relative to uniform@8.

Policy	Setting	$\bar{b}$	Acc.	Δ	Degrades
Uniform	$b = 7$	7.00	74.76	-0.40	n/a
Uniform	$b = 8$	8.00	75.16	$+0.00$	n/a
Uniform	$b = 10$	10.00	75.52	$+0.36$	n/a
METAGATE	$\tau = 0.3$	6.86	75.49	$+0.33$	1/20
METAGATE	$\tau = 0.5$	6.99	75.44	$+0.28$	1/20
METAGATE	$\tau = 0.7$	7.21	75.39	$+0.23$	0/20

The GSM8K transfer matrix is less discriminative because the models are near ceiling. There, ANLL adaptive allocation averages $+0.31$ pp over uniform across the 20 transfers, while METAGATE is essentially accuracy-neutral (-0.01 pp) at a lower average budget of 6.52 samples. We therefore use MATH-500 as the main stress test and GSM8K to measure accuracy retention in a high-accuracy regime.

Phi cross-family check. We add Phi-4-mini-instruct as a non-Qwen target/source on MATH-500. Qwen→Phi transfers favor adaptive ANLL, whereas Phi→Qwen transfers produce a pronounced transfer asymmetry. Over all 10 transfer pairs involving Phi, ANLL averages -0.42 pp relative to uniform and degrades in 5/10 pairs; METAGATE averages $+0.16$ pp, degrades in 2/10 pairs, uses 7.56 decoded samples, and beats matched-budget random fallback by $+0.61$ pp (pair-bootstrap 95% CI $[+0.19, +1.07]$). In the Phi→Qwen direction, ANLL degrades in all 5 pairs with mean -1.72 pp; METAGATE reduces this to one below-uniform cell, averages $+0.00$ pp at 6.23 samples, and beats matched-budget random by $+1.06$ pp. Thus Phi provides cross-family corroboration of the downside-control effect.

Table 2 checks the self-baseline behavior on the settings used in the original evaluation. METAGATE matches uniform accuracy to within 0.2 pp in all five settings and exceeds it by 0.2 pp on 8B and 30B MATH, while the adaptive allocator is below uniform on three of five self-baseline settings (8B MATH raw paired McNemar $p=0.035$). It does so at $\bar{b}=6.2$ to 7.0 samples versus uniform’s 8.0, a 12 to 22% decoded-sample budget saving. Because the gated budget is not pinned to B , we report realized budget rather than assume compute parity.

Table 2. Self-baseline accuracy (%) and realized budget $\bar{b}$ (samples), with $B = 8$, $k = 4$, $N = 16$. Acc@16 is the maximum-sample reference. Bold: best among Uniform, Adaptive, MetaGate; †: adaptive below uniform.

	MATH-500 ($n=500$)			GSM8K ($n=1319$)	
Method	8B	14B	30B	8B	14B
Uniform	77.2	77.2	77.0	90.9	94.5
Adaptive	†75.4	†77.0	†76.0	91.0	94.5
MetaGate	77.4	77.2	77.2	90.8	94.5
Acc@16 (ref.)	77.2	77.8	77.4	91.4	94.3
$\bar{b}$ Adaptive	8.01	8.01	8.01	8.00	8.00
$\bar{b}$ MetaGate	7.04	6.83	7.04	6.24	6.34

For 8B MATH, Acc@16 and uniform@8 are both 77.2%. Its -1.8 pp adaptive gap therefore combines signal ranking with the absence of an aggregate gain from additional samples in this setting. By contrast, Acc@16 is 0.6 and 0.4 pp above uniform@8 for 14B and 30B MATH, respectively.

Statistical testing. The self-baseline tests support the same reading as the transfer audit: adaptive allocation has no stable advantage over uniform. The largest gap is on 8B MATH ($\Delta = -1.8$ pp, paired McNemar $p = 0.035$); the other four settings are mixed ($p > 0.3$). Policy comparisons use paired McNemar tests and Holm correction; signal correlations use bootstrap CIs. Because transfer cells reuse both deployment models and query identities, the main macro-policy intervals resample the 500 query identities while preserving their repeated use across all 20 off-diagonal pairs. Leave-one-target-out estimates keep the MetaGate mean between $+0.23$ and $+0.34$ pp, whereas ANLL remains between -0.69

and -0.26 pp. We reserve pair-bootstrap intervals for descriptive pair-level controls; the primary intervals use the query-clustered analysis above.

Operationally, adaptive uses budgets in $\{4, 16\}$, whereas MetaGate adds the uniform fallback $B=8$ for low-confidence queries.

5.4 Additional Selectivity Analysis

If MetaGate only reduced adaptive usage, random fallback with the same coverage or realized budget should perform similarly. We test both comparisons directly.

Ranking harmful decisions (AUROC). We label a query *harmful* if adaptive gets it wrong while uniform gets it right. Pooling all 20 transfer pairs (10,000 query-instances, 181 harmful, base rate 1.8%), we ask whether low meta-confidence ranks these harmful cases first. Using $1 - m(x)$ as the score, so higher means lower meta-confidence, we obtain pooled AUROC = 0.809. To avoid relying only on pooled pair-level variation, we also compute AUROC within each transfer pair and then macro-average: pair-macro AUROC is 0.799 with bootstrap 95% CI [0.748, 0.847]. For these rare events, pooled AUPRC is 0.060 at a 0.018 base rate, and pair-macro AUPRC is 0.090 with 95% CI [0.066, 0.114], confirming useful ordering beyond the base rate. At the fixed operating point $\tau=0.5$, the fallback branch covers 50.5% of query-instances and captures 91.7% of harmful adaptive decisions, with harmful precision 3.3% (macro recall 91.4%, macro precision 3.3%). This matches the design goal: recall downside cases and use uniform fallback, rather than abstention or human escalation. The fallback branch uses the ordinary $B=8$ vote, while the overall MATH transfer budget at $\tau=0.5$ remains 6.99 samples because trusted low-risk queries often stop at $k=4$. We use the risk versus coverage and random-fallback controls below as policy-level evidence.

Risk versus coverage vs. random fallback. As an equal-coverage stress check, on the degrading 14B $\rightarrow$ 8B pair (uniform 77.2, full-adaptive 75.0), meta-confidence-ranked trust beats random trust by 0.6 to 1.8 pp across nontrivial coverages and by 1.3 to 1.5 pp in the 40 to 60% operating range (e.g., 77.6 vs. 76.3 at 40% coverage and 77.4 vs. 75.9 at 60%). Equal-fraction random gating therefore recovers far less of the downside, pointing to *selectivity* beyond reduced adaptive use.

Matched-budget random fallback. Equal fallback rate alone does not guarantee equal cost: the trusted subset can contain different fractions of high-budget queries. We therefore add a stricter random control that matches both the number of trusted high-budget and trusted low-budget queries, and hence matches the realized sample budget. Across all 20 off-diagonal transfers, METAGATE beats this matched-budget random control by +0.79 pp on average (pair-bootstrap 95% CI [+0.63, +0.96]); the gap is +0.86 pp on ANLL-degrading transfers and +0.65 pp on the remaining transfers. Table 3 reports all 13 MATH

transfer pairs where ANLL adaptive allocation is below uniform. METAGATE is at least as accurate as uniform in 12 of 13, with one -0.2 pp cell.

Table 3. All MATH-500 transfer pairs where ANLL adaptive allocation degrades below uniform. Random controls match METAGATE’s realized sample budget; values are accuracy (%) over 1000 random trials.

Transfer	Uniform	Adaptive	MetaGate	$\bar{b}_{\text{MG}}$	Matched random
1.7B→8B	77.2	76.4	77.6	7.47	76.4±0.4
1.7B→30B	77.0	76.6	76.8	7.07	76.6±0.3
4B→1.7B	70.0	69.6	70.2	6.92	69.5±0.4
4B→8B	77.2	76.4	77.2	7.33	76.4±0.3
4B→30B	77.0	76.4	77.0	7.00	76.5±0.3
8B→1.7B	70.0	68.6	70.2	6.74	69.1±0.5
8B→14B	77.2	77.0	77.6	7.07	76.9±0.3
8B→30B	77.0	76.2	77.2	6.78	76.4±0.3
14B→1.7B	70.0	68.0	70.2	6.85	69.0±0.5
14B→8B	77.2	75.0	77.4	6.77	75.7±0.4
14B→30B	77.0	76.2	77.2	6.59	76.5±0.3
30B→1.7B	70.0	69.6	70.2	6.93	69.6±0.4
30B→8B	77.2	76.4	77.4	7.17	76.4±0.3

Representative gate decisions. Table 4 traces three 14B→8B queries. Proxy disagreement correctly triggers fallback on the modular-inverse problem, while proxy agreement preserves a useful high-budget decision on the tangent problem. The parallel-lines example complements these successes with a boundary case in which proxy agreement preserves an incorrect adaptive vote.

Table 4. Representative MATH-500 decisions for 14B→8B. Signal values are source-z-scores; answers show uniform@8 versus the trusted or proposed adaptive branch.

Case	$\hat{s}_{\text{lp}}$	$\hat{s}_{\text{cons}}$	$m(x)$	Decision and outcome
Modular inverse	-0.53	+2.40	0.00	fallback: 17 correct vs. adaptive@4 19
Tangent angles	+1.28	+1.38	0.98	trust: adaptive@16 180° correct vs. 90°
Parallel lines	+1.97	+2.40	0.70	trust: adaptive@16 124° wrong vs. 28°

5.5 Experiment 3: Ablations

Held-out source calibration. To check that source-query calibration is not driving the result, we repeat the MATH-500 audit with 50 random 50/50 source splits: thresholds and z-score statistics are fit on one half and policies evaluated on the held-out half. Across off-diagonal transfers, adaptive ANLL has macro $\Delta = -0.58$ pp (95% split interval $[-1.31, +0.24]$), while METAGATE has $\Delta = +0.26$ pp ($[+0.04, +0.46]$) at mean budget 7.02.

Table 5. Compact sensitivity check on 1.7B→8B MATH at $\tau=0.5$.

k	1	2	4	6	8
Acc.	76.4	77.0	77.6	77.2	77.2
$\bar{b}$	8.81	6.79	7.47	7.90	8.00

α	0.5	1	2	4	8
Acc.	77.2	77.4	77.6	77.2	77.4
Trust	70%	56%	42%	31%	24%

Threshold and hyperparameter sensitivity. On 8B MATH, increasing τ monotonically shrinks the trusted fraction and recovers uniform at $\tau > 1$; the operating range $\tau \approx 0.4$ to 0.6 matches uniform while trusting roughly 40 to 60% of queries. Table 5 gives a compact single-pair sensitivity check on 1.7B→8B MATH: the tested $\alpha \in \{0.5, 1, 2, 4, 8\}$ sweep stays within 0.4 pp, while $k \in \{1, 2, 4, 6, 8\}$ varies by up to 1.2 pp because it changes both minimum budget and the consistency estimate. We use $k=4, \alpha=2.0$ in the main experiments and restrict $k \leq B$.

Stronger base signals. We also test answer consistency ($1 - c(x)$) and answer-distribution entropy from the $k=4$ probes. Both improve over ANLL (Table 6). With only the trusted-branch allocator replaced, the unchanged gate moves the $B=8$ operating point from +0.66 pp at 8.33 samples to +0.31 pp at 7.20 samples while retaining the same 1/20 downside count. Thus consistency and gated consistency form complementary operating points: +0.66 pp at 8.33 samples for accuracy, and +0.31 pp at 7.20 samples for lower decoded-sample cost with the same downside count.

Table 6. Base-signal and fixed-gate comparison over the 20 off-diagonal MATH-500 transfers. Each Δ is relative to uniform at the matching nominal B ; $\bar{b}$ is the realized target budget. “+MG” keeps the ANLL-consistency meta-confidence and all gate parameters unchanged.

Policy	B	Δ (pp)	Degrades	$\bar{b}$
ANLL	8	−0.44	13/20	8.03
ANLL + MG	8	+0.28	1/20	6.99
Consistency ($k=4$)	7	+1.01	0/20	7.75
Consistency ($k=4$)	8	+0.66	1/20	8.33
Consistency + MG	8	+0.31	1/20	7.20
Entropy ($k=4$)	7	+0.96	0/20	7.69
Entropy ($k=4$)	8	+0.66	1/20	8.33
Entropy + MG	8	+0.31	1/20	7.20

6 Conclusion

This paper establishes deployment-target dependence as a central factor in frozen ANLL allocation. With predeclared hyperparameters, METAGATE’s label-free fallback averages +0.28 pp over uniform under a query-clustered analysis, uses 6.99 samples, and has one observed below-uniform transfer; its advantage over budget-matched random fallback supports selectivity. The selective improvement criterion characterizes how trusted adaptive decisions and uniform fallback

jointly determine risk. Consistency and gated consistency provide complementary operating points: +0.66 pp at 8.33 samples for accuracy, and +0.31 pp at 7.20 samples for lower decoded-sample cost with the same downside count.

Limitations. Our evaluation covers sampling-based TTC on Qwen3 and Phi models across MATH-500 and GSM8K. Query-clustered inference and leave-one-target-out analysis account for shared queries and deployment targets, while the fixed $\tau = 0.5$ protocol isolates label-free transfer. These results provide a foundation for evaluating agreement gating with independent decoding pools and additional reasoning workloads.

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